



Continuity Models

2.4.1. Continuity Models

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What is a Continuity Model?

Continuity Models: Basic Definitions

- **What is a Continuity Model?**

Sequence of algorithms/rules to identify **continuity states** from each user's individual events.

- **What is a Continuity State?**

Continuity states describe a **person's mobility condition** at a given moment.

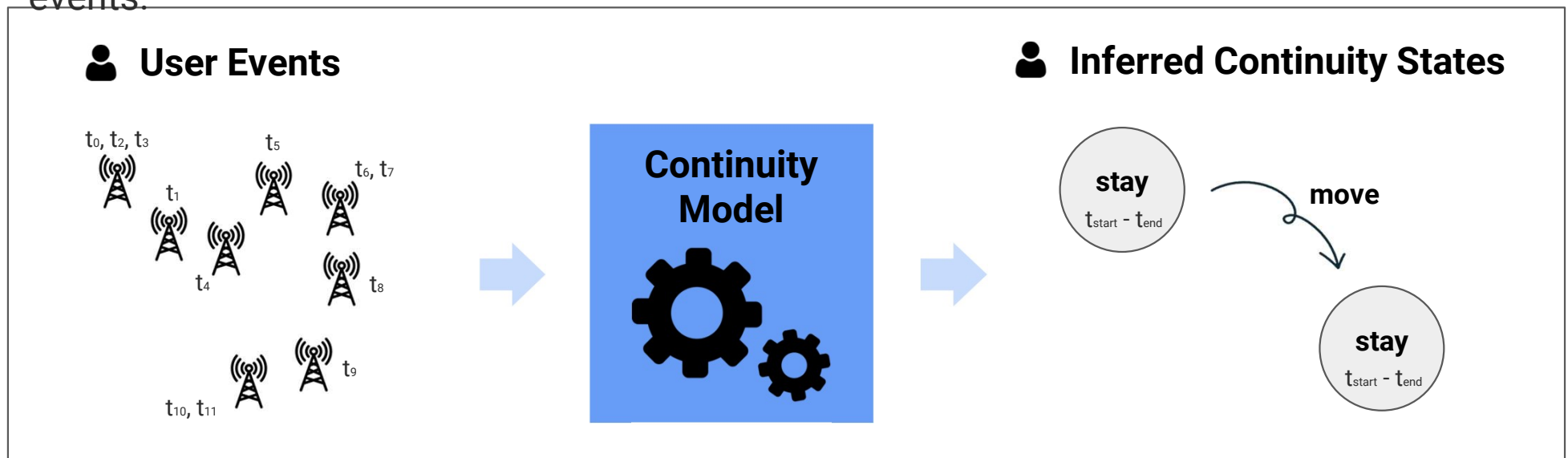
Their **exact definition** is **highly use case dependent**, but there are **two main states**:

- **Stay**: the person remains in the same location without significant movement.
- **Move**: the person is changing location, indicating displacement or travel.

Continuity Models: Basic Definitions

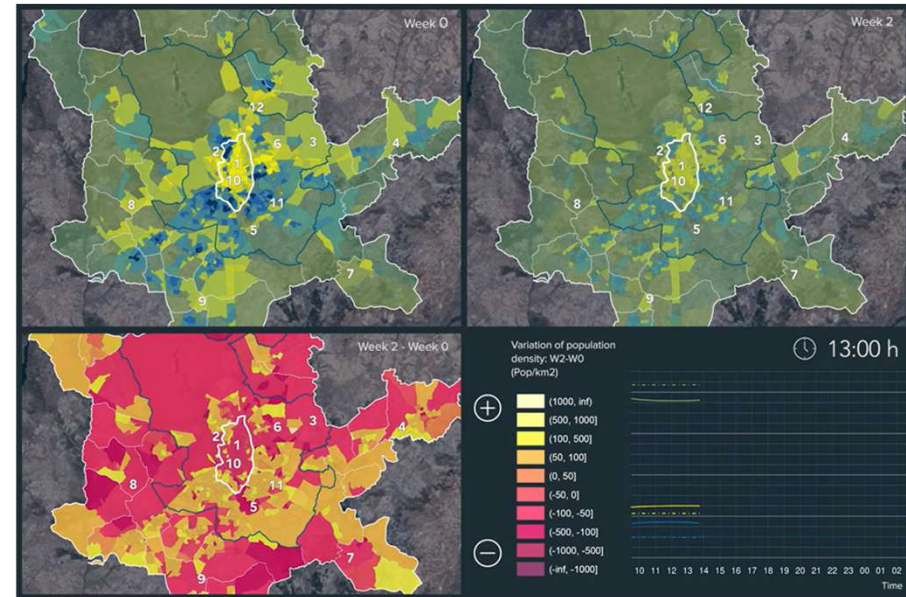
- What is a Continuity Model?

Sequence of algorithms/rules to identify **continuity states** from each user's individual events.



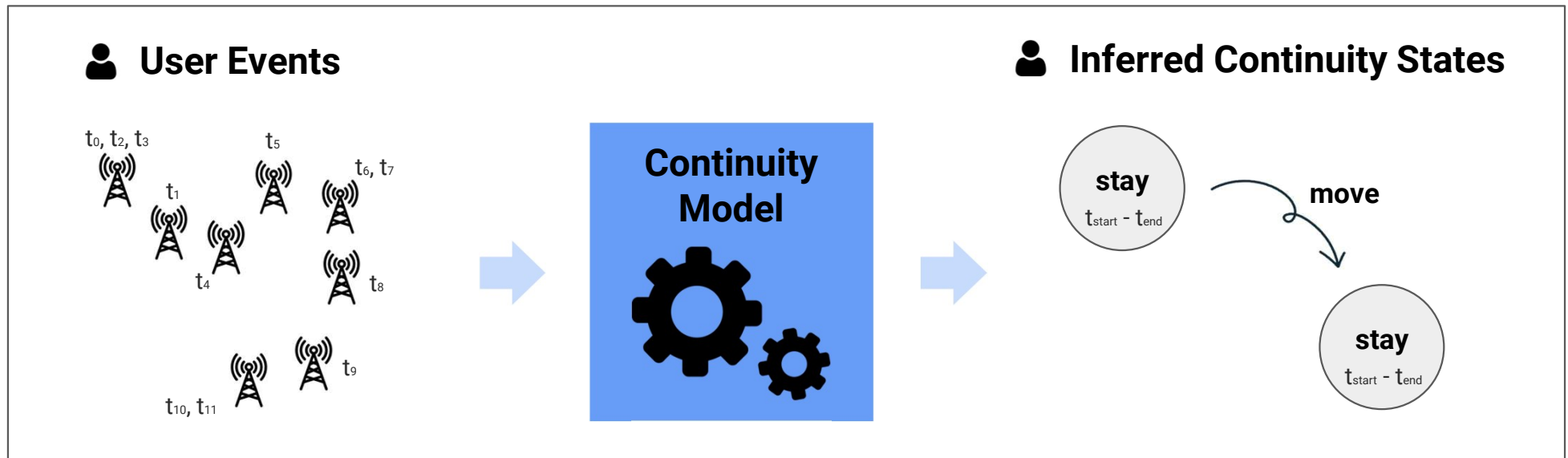
Why do we need Continuity Models?

- Most Mobile Phone Data (MPD) analyses need users' continuity states:
 - **Travel and commuting:** identify moves for each user, with origins, destinations, and times, to build OD matrices and trip statistics.
 - **Tourism:** identify stays to calculate PoI visit duration, most visited locations and other location-based indicators.
 - **Disaster management:** determine where people actually stay to support epidemic control and emergency planning.
 - Etc.



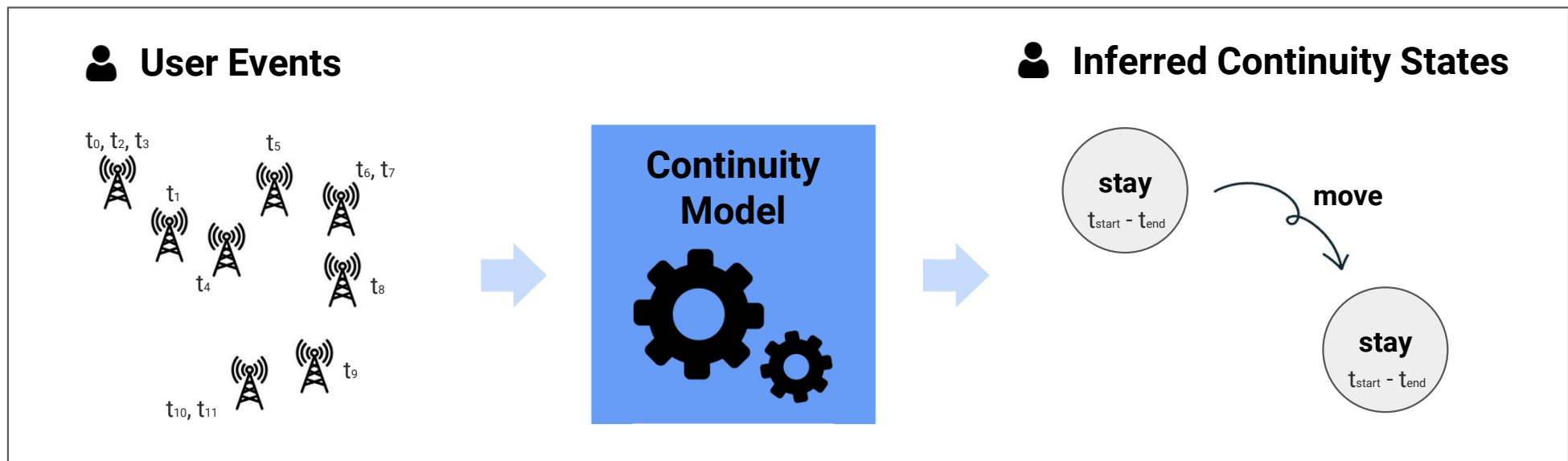
Dynamic population maps for the Madrid during the COVID-19 pandemic lockdown, by Gustavo Romanillos, tGIS Research Group, Complutense University of Madrid, powered by NOMMON.

Why are Continuity Models important?



Sounds easy so...
What is the challenge?

Why are Continuity Models important?

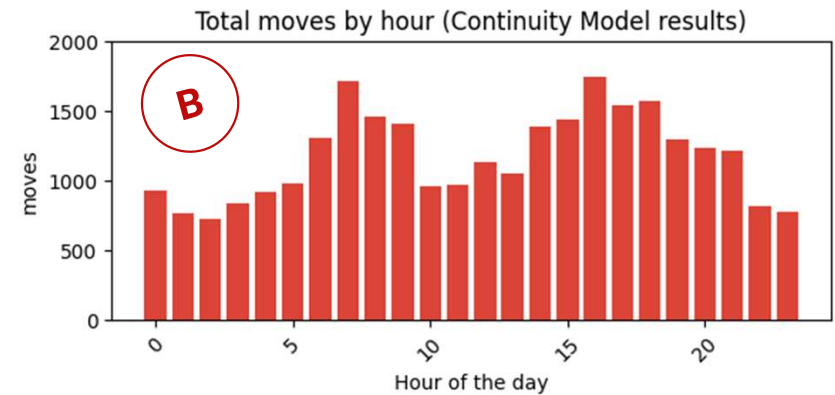
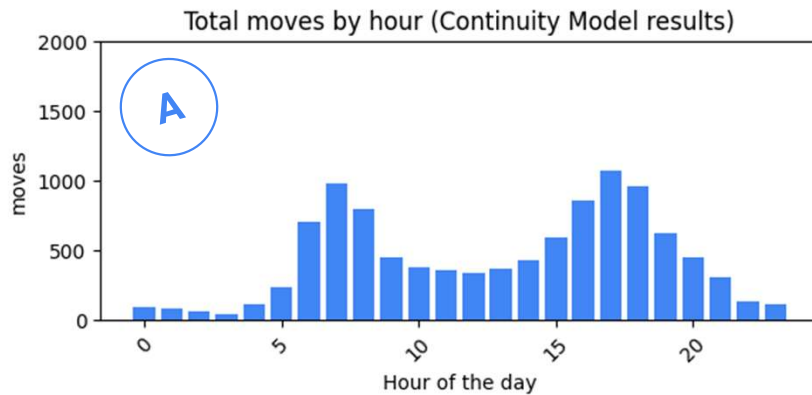


*Sounds easy so...
What is the challenge?*

! *In MPD, not every cell change means movement, and not every repeated cell event implies a stay!*

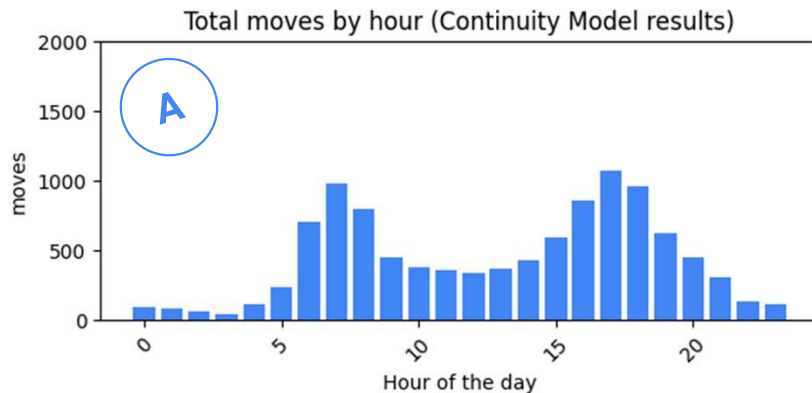
Why are Continuity Models important?

- The consequences of an oversimplified Continuity Model

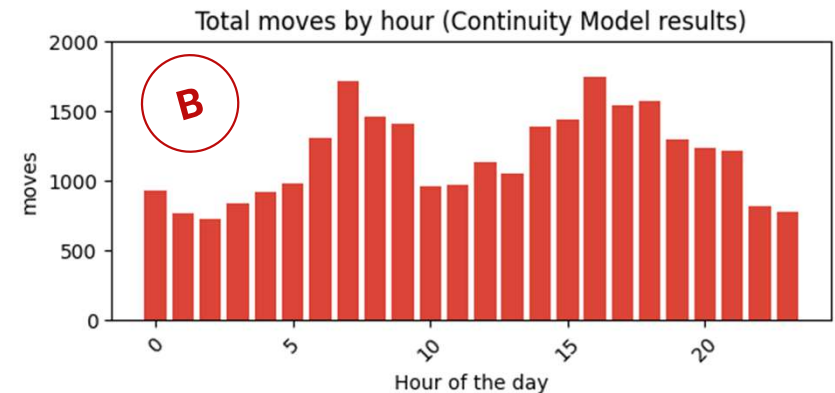


Why are Continuity Models important?

- The consequences of an oversimplified Continuity Model



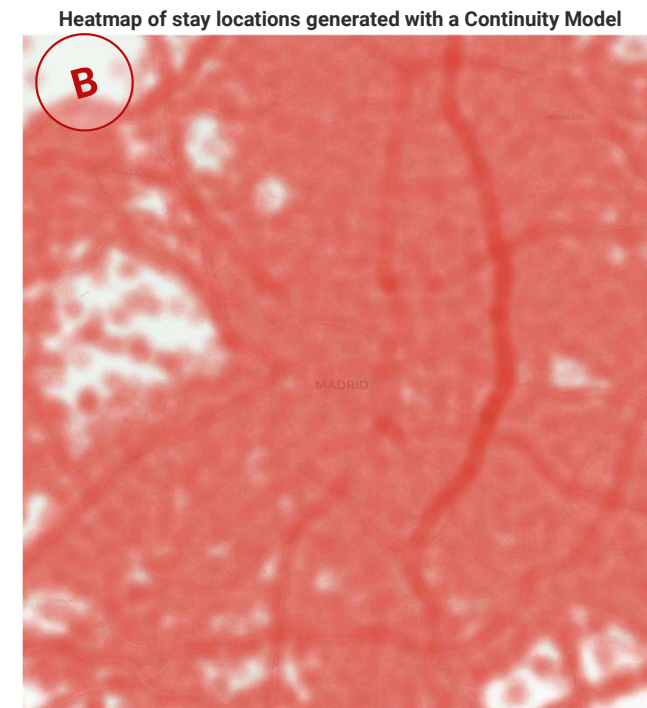
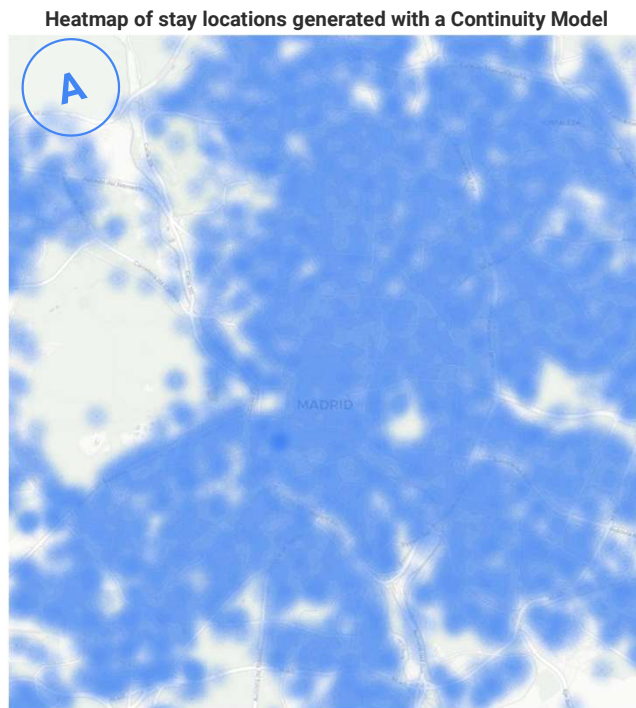
A **robust continuity model** evaluates cell changes in context and only flags a move when the evidence is clear. It separates real movements from network noise in most cases, resulting in reliable hourly move volumes, e.g. showing realistic night-time activity.



A **naive continuity model** that treats every cell change as a move generally overestimates moves. It detects real movements, but also many false positives. The resulting hourly move volumes show strong base noise, e.g. clearly inflating night-time trips.

Why are Continuity Models important?

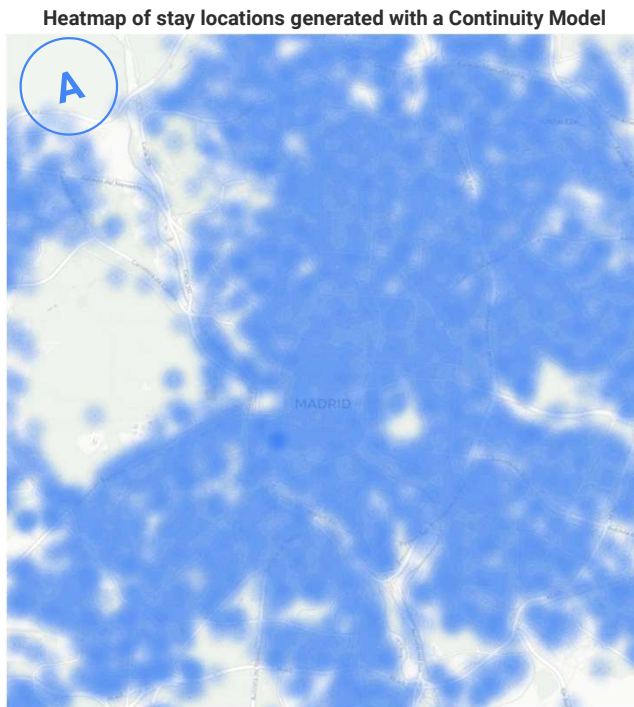
- The consequences of an oversimplified Continuity Model



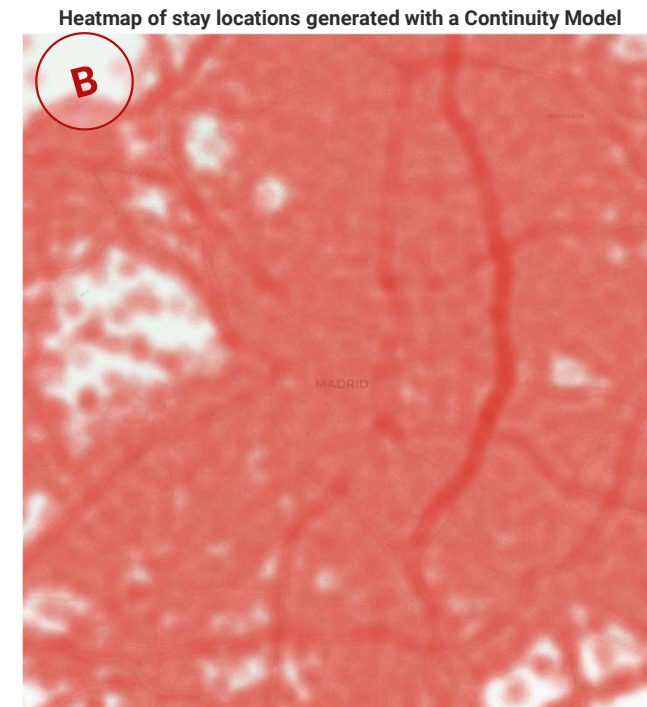
Why are Continuity Models important?

- The consequences of an oversimplified Continuity Model

A **robust continuity model** produces stays that reflect real behaviour. The resulting stay location heatmaps highlight areas where users actually remained for meaningful time, including residential, office, leisure, touristic zones, etc.

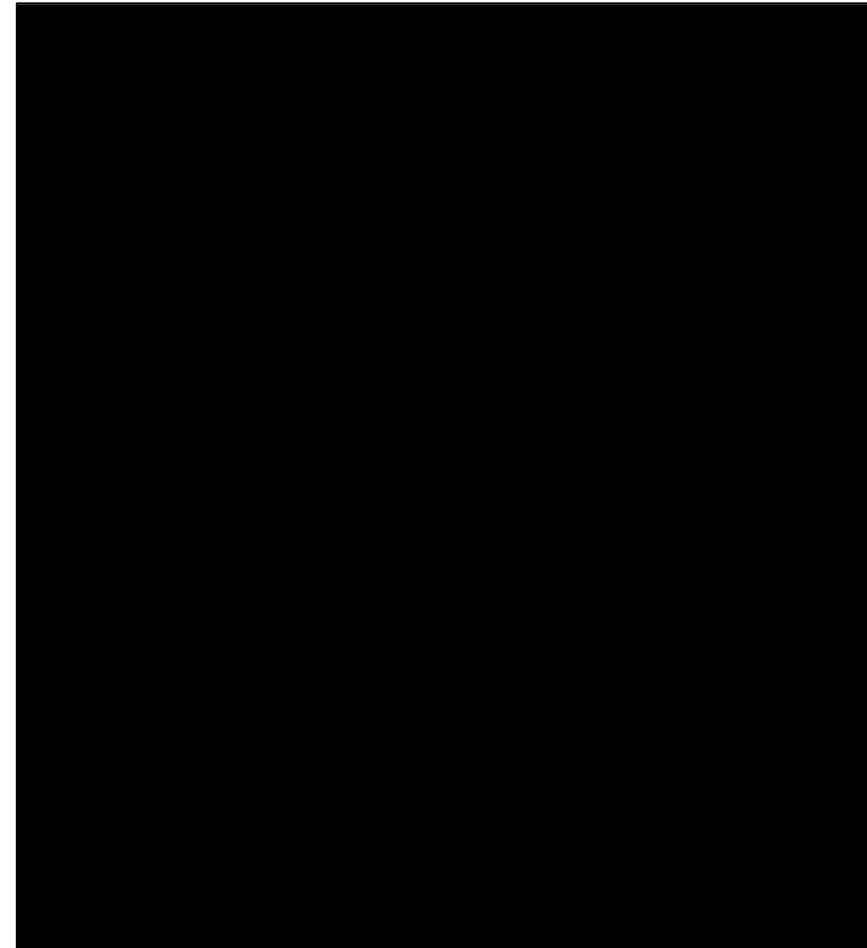


A **naive continuity model** produces stay locations that generally do not align with the expected results. Heatmaps generated with their output favour locations that were not truly visited but simply passed through, such as main roads.



Let's play Continuity Model

- *How many different stays does this user have during this day? Where and when?*



 User Events

[FLOWMINDER.ORG](https://flowminder.org)

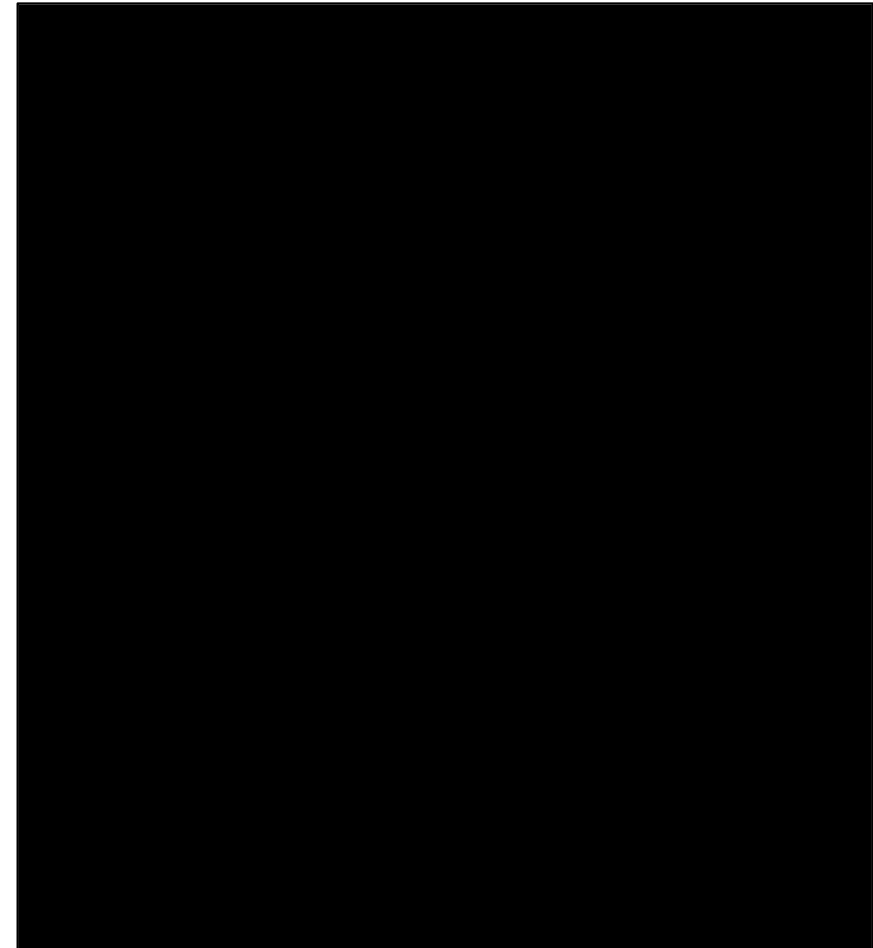
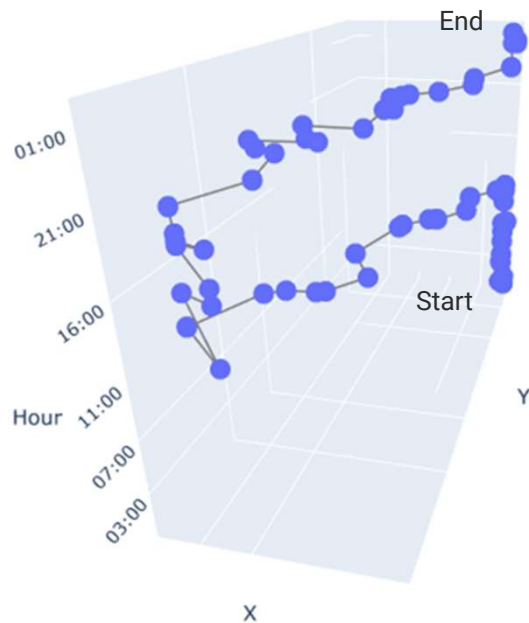
[NOMMON](https://nommon.org)

 positium

Let's play Continuity Model

- *How many different stays does this user have during this day? Where and when?*

Using time on the z-axis in a 3D plot allows visualising the user's full event-location sequence at one glance.



Let's play Continuity Model

- *How many different stays does this user have during this day? Where and when?*

3-4
stays

Using time on the z-axis in a 3D plot allows visualising the user's full event-location sequence at one glance.



Let's play Continuity Model

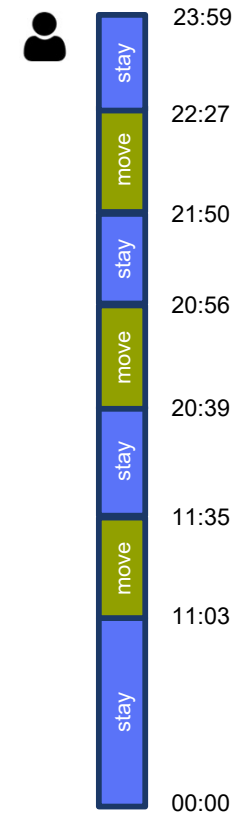
- How many different stays does this user have during this day? Where and when?

4
stays

Using time on the z-axis in a 3D plot allows visualising the user's full event-location sequence at one glance.



Continuity Model Results:



MPD Continuity Models: Initial considerations

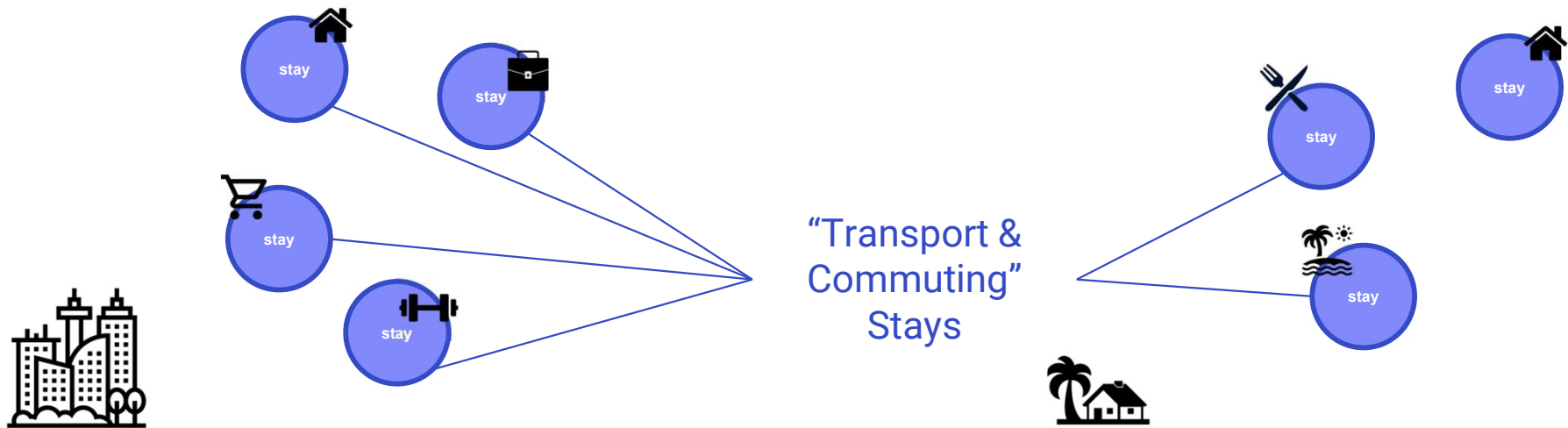
There is no one-size-fits all

- We may find **many differences** in Mobile Phone Data (MPD) analyses:
 - **Data provider heterogeneity:** event granularity and data fields vary across telecom operators.
 - **Variability within the same raw data:** within a single MPD dataset, some devices generate many events while others produce only a few. Antenna density also differs across areas (urban/rural).
 - **Use Cases:** stay definitions and required precision may vary depending on the purpose (e.g., migration vs. commuting) and spatio-temporal scope.



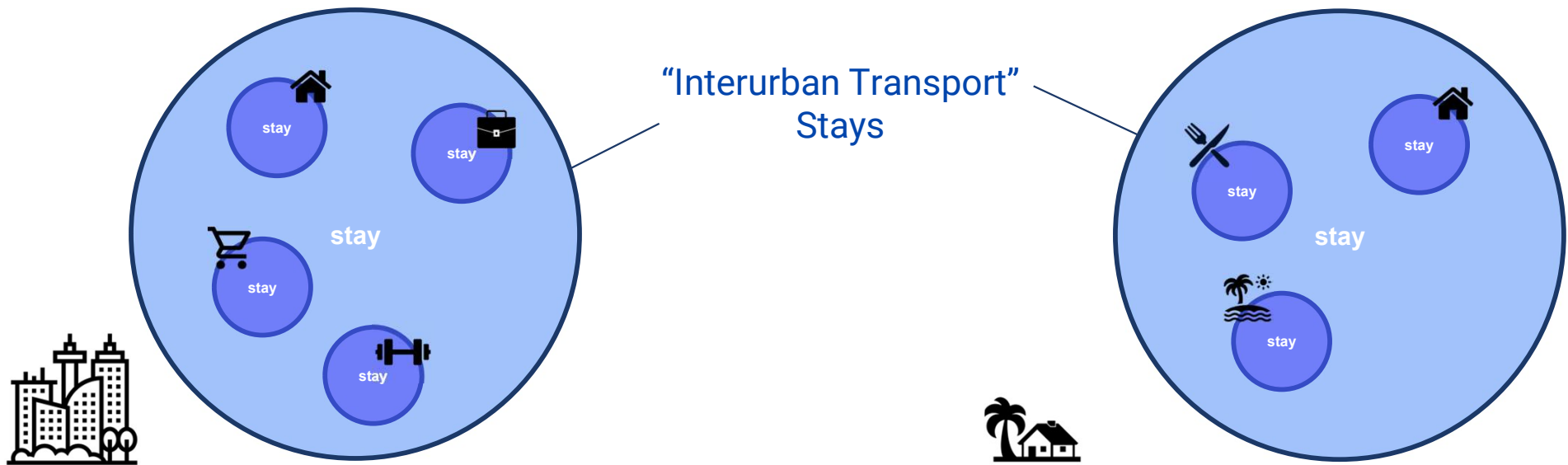
There is no one-size-fits all

- Spatiotemporal levels for what a **stay**/move is are **Use Case dependent**



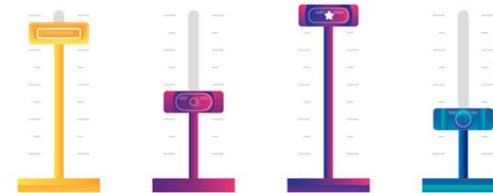
There is no one-size-fits all

- **Spatiotemporal levels** for what a **stay/move** is are **Use Case dependent**



There is no one-size-fits all

- These differences require **flexible** Continuity Model designs. Specifically:
 - An ideal continuity model should be **configurable**, with modifiable spatial and temporal parameters that can be tuned to different contexts (data sources, use cases, scopes, etc.).
 - An ideal continuity model should be **adaptive**, so that within each analysis it can dynamically adjust to handle uneven event density across areas and devices.



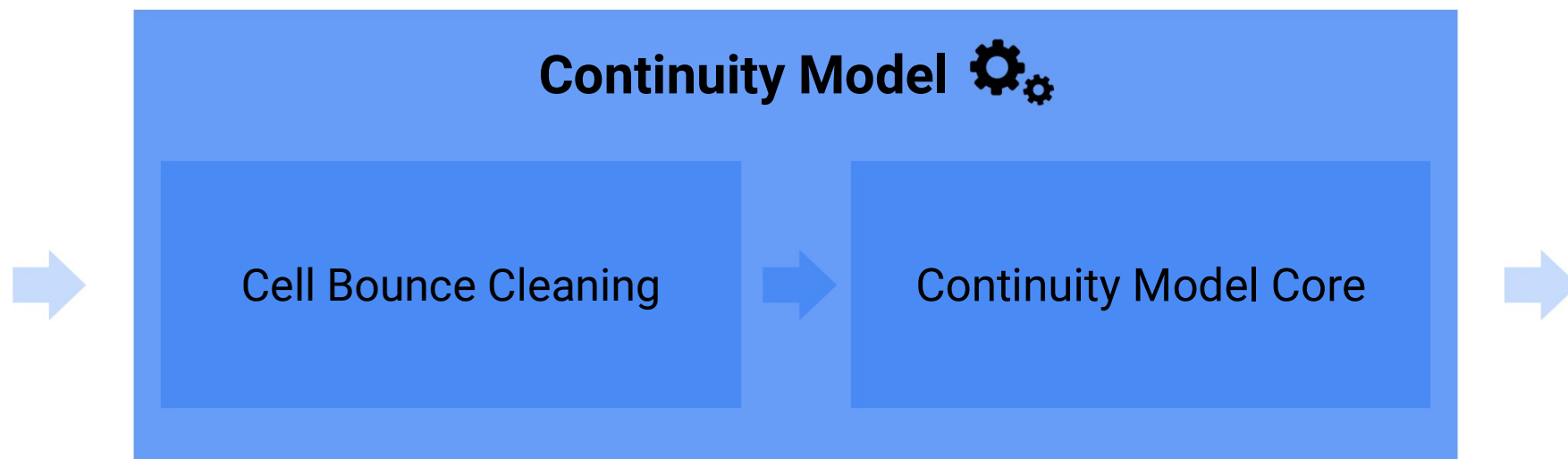
Calibration & Validation



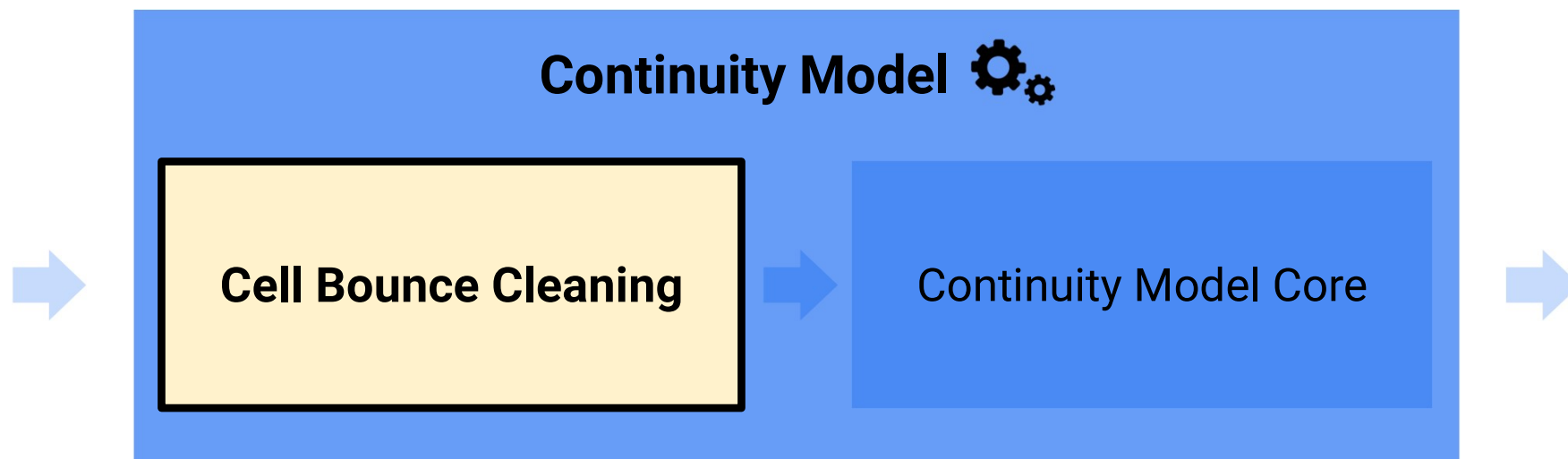
- For any Continuity Model, it is essential to ensure that it accurately captures the real movement patterns of the users.
- To achieve this, we can rely on:
 - **User-level reported mobility** (e.g. from users that voluntarily agree to share their movements through surveys or apps) is ideal if available, since it allows to validate model outputs against actual user behavior. Direct accuracy metrics (precision/recall) can be used for calibration.
 - **Visual inspection** of user trajectories is useful for checking coherence in stay/move labeling.
 - **External data sources** such as surveys (e.g., trips per person/day) can also be very useful for parameter calibration by validating at aggregated level.

MPD Continuity Models: Cell Bounce Cleaning

Cell Bounce Cleaning for Continuity Models



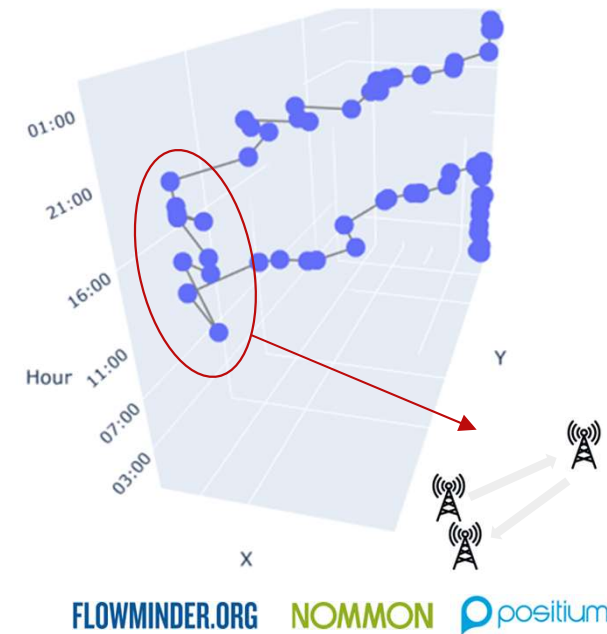
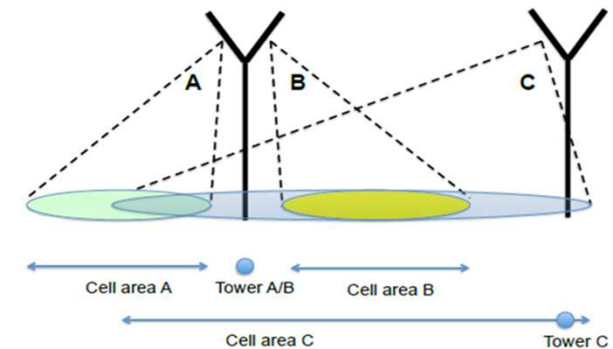
Cell Bounce Cleaning for Continuity Models



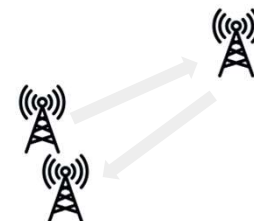
Cell Bounce Cleaning

! The “cell bouncing” problem

- In Mobile Phone Data (MPD), cell **coverage areas often overlap**, meaning a device at a stationary location can connect to two or more cells depending on network conditions or signal strength.
- As a result, a cell change does not necessarily indicate movement.
- In fact, very often in MPD users may appear to “**bounce**” **between neighboring cells** (“cell bouncing” or “ping-pong handovers”) even when stationary.



Cell Bounce Cleaning

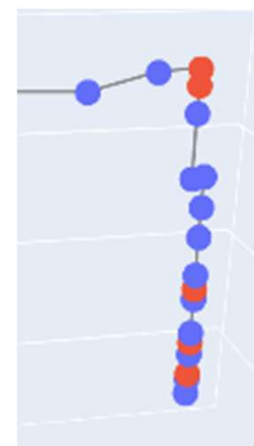
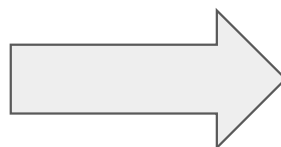
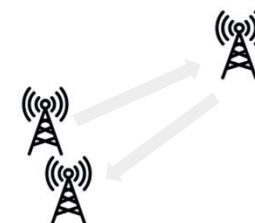


💡 Cell bounce cleaning approach

- This is “optional”, but skipping this step can lead to an **overestimation of moves**.
- There are many possible approaches (some more complex than others). A **simple but effective method** can consist of:
 - Ignoring **quick switches between neighboring cells** followed by a return.
 - For example, we can apply a **30-minute threshold**: if a user connects to **cell A** → **other cells** → **back to A**, all within less than 30 minutes, we can assume the intermediate events are noise or cell bounces and either discard them or replace their cell ID with A.

Cell Bounce Cleaning

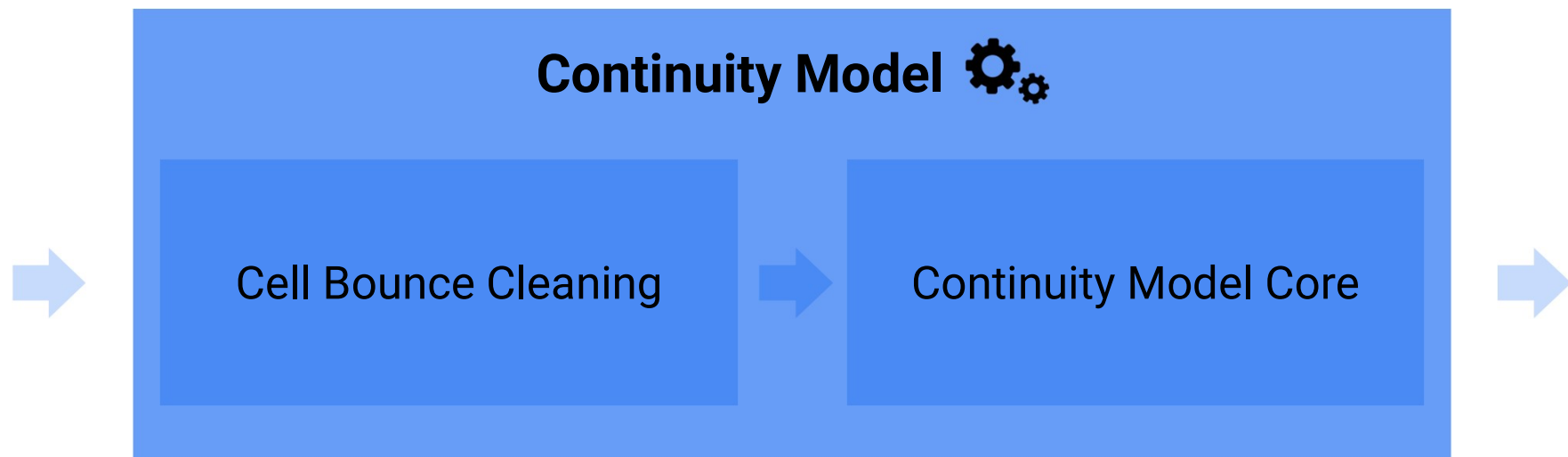
💡 Cell bounce cleaning approach



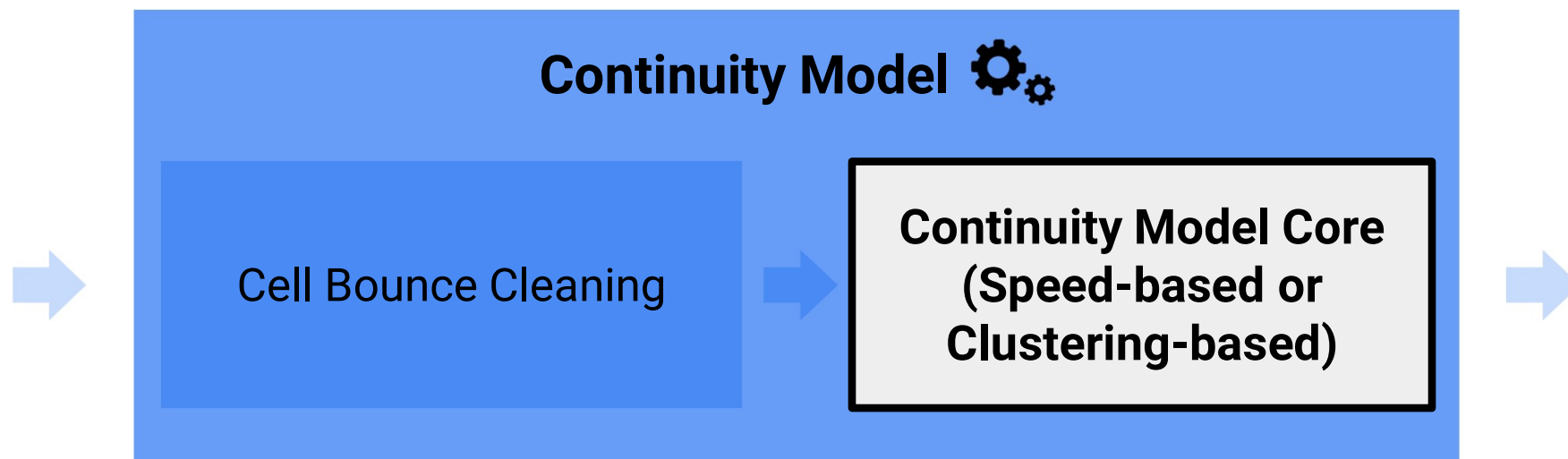
preprocessed
● False
● True

MPD Continuity Models: Core Methods

MPD Continuity Models Core

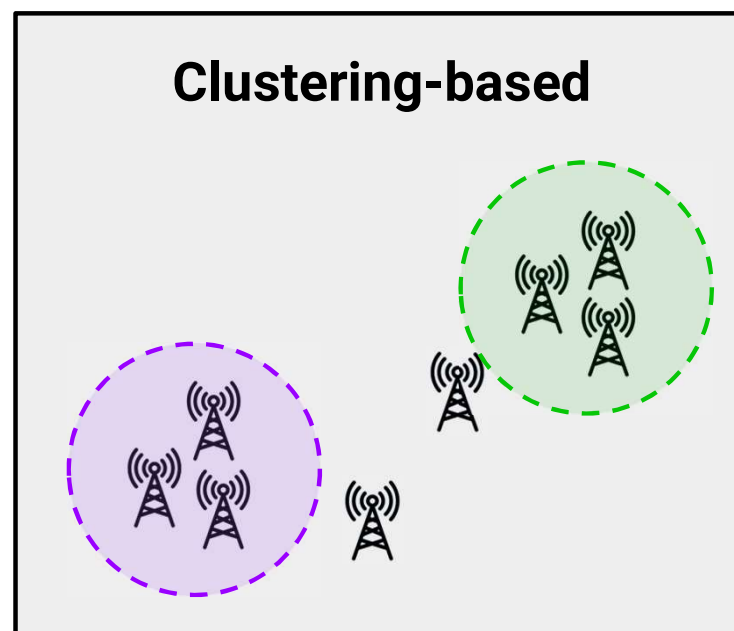
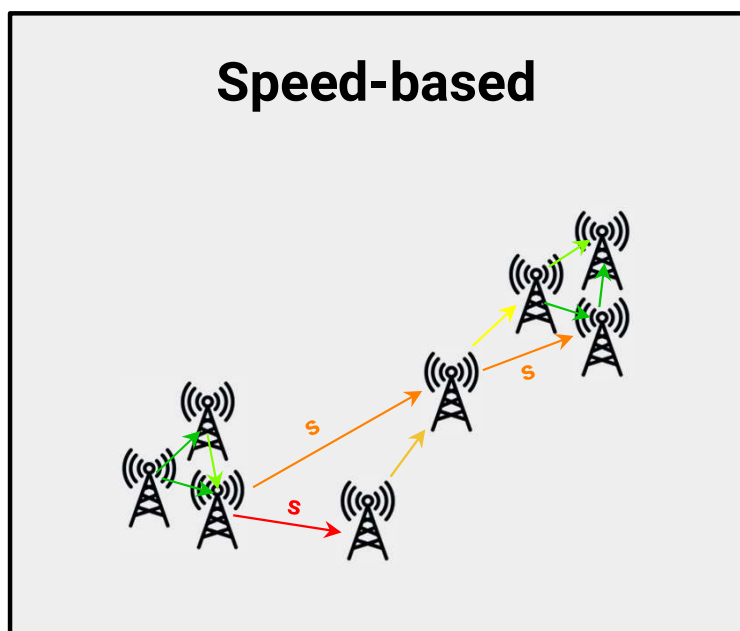


MPD Continuity Models Core



MPD Continuity Models Core: Basic approaches

Two basic approaches for stay/move differentiation:



Speed-Based

- **Key idea:** Comparing **speeds** of **event sequences** to a **speed threshold** (e.g. 10 km/h).
- Ideally, these thresholds should be **configurable** and **adaptable**.
- *E.g.: calculating speeds for $A1 \rightarrow A2$, $A1 \rightarrow A2 \rightarrow A3$ and $A2 \rightarrow A3$ event sequences by dividing cell location distances by the timestamp differences, then comparing to some dynamic speed thresholds depending on the device and region.*

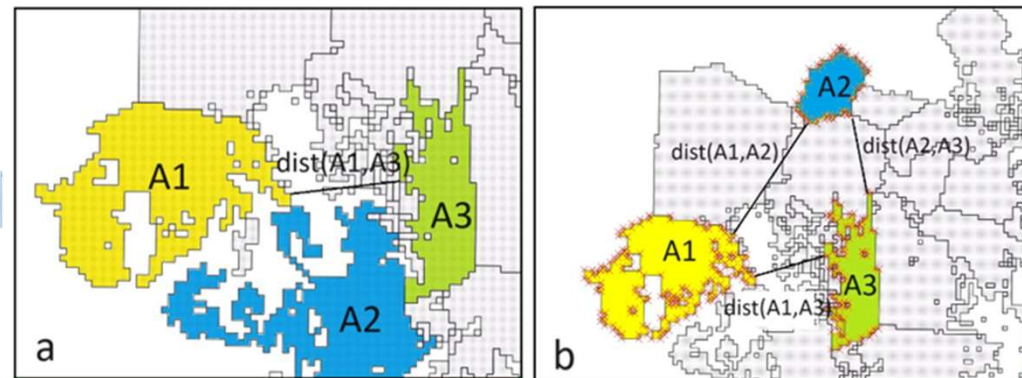
A methodological example of a speed-based continuity model with square grid cell coverage areas from Eurostat MultiMNO project (see section “14.2 METHOD 2: DAILY PERMANENCE SCORE ESTIMATION”):

[D2.3 – Volume III – Methods and data objects](#)

Multi-MNO project



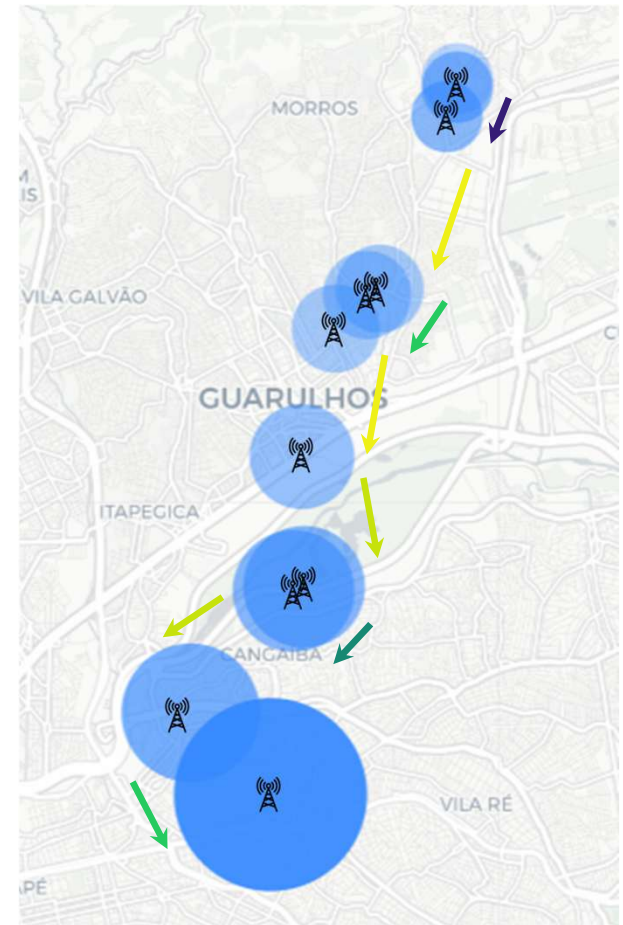
Eurostat CROS



Speed-Based

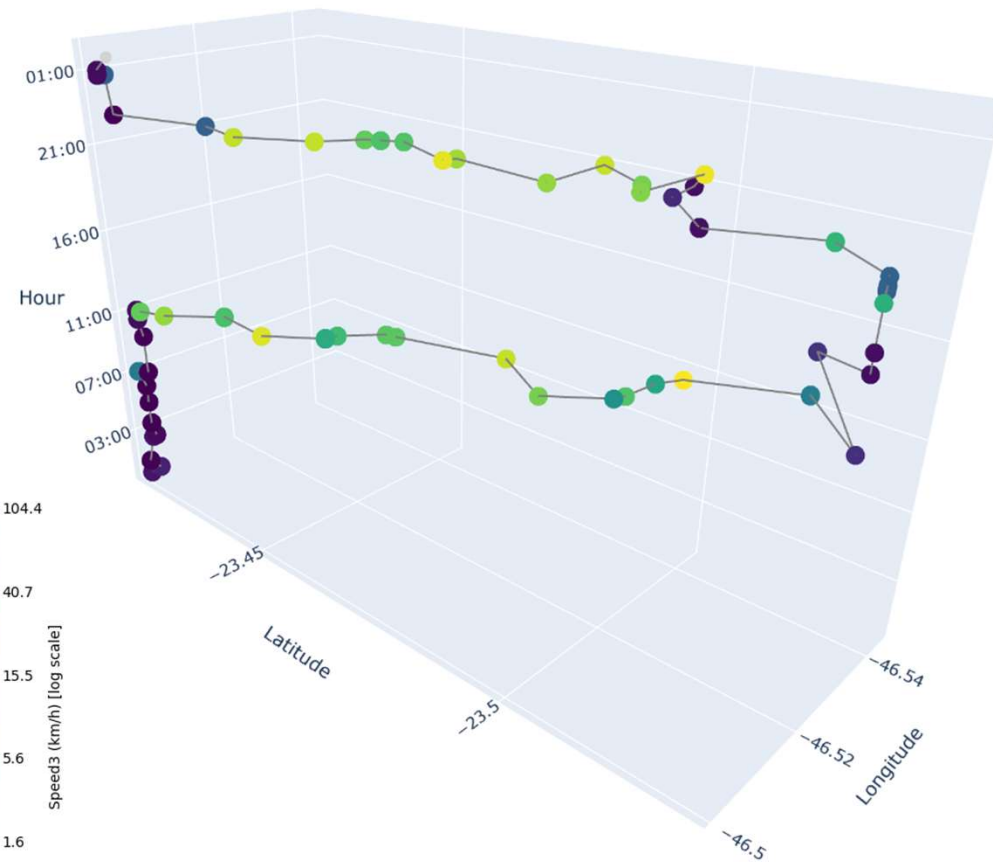
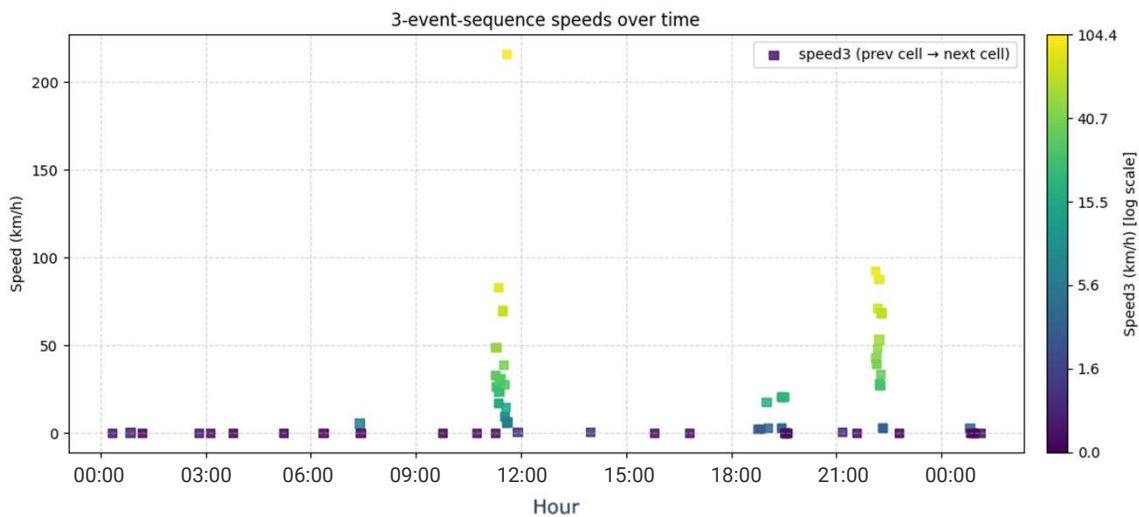
- A **simple approach** can consist on **calculating speeds between the first event at a given cell A and the first event at a different cell B** and assigning this speed to all events between them.

$$\text{Speed}(A, B) = \text{TimeDifference}(A, B) / \text{TravelDist}(A, B)$$



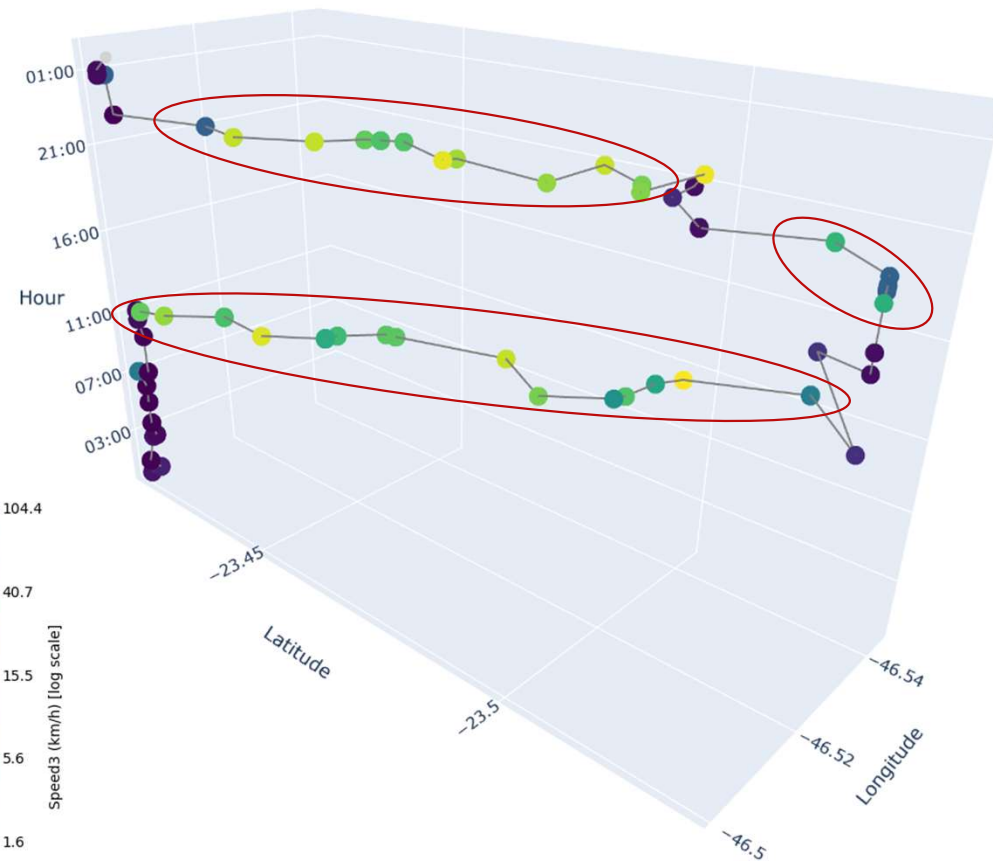
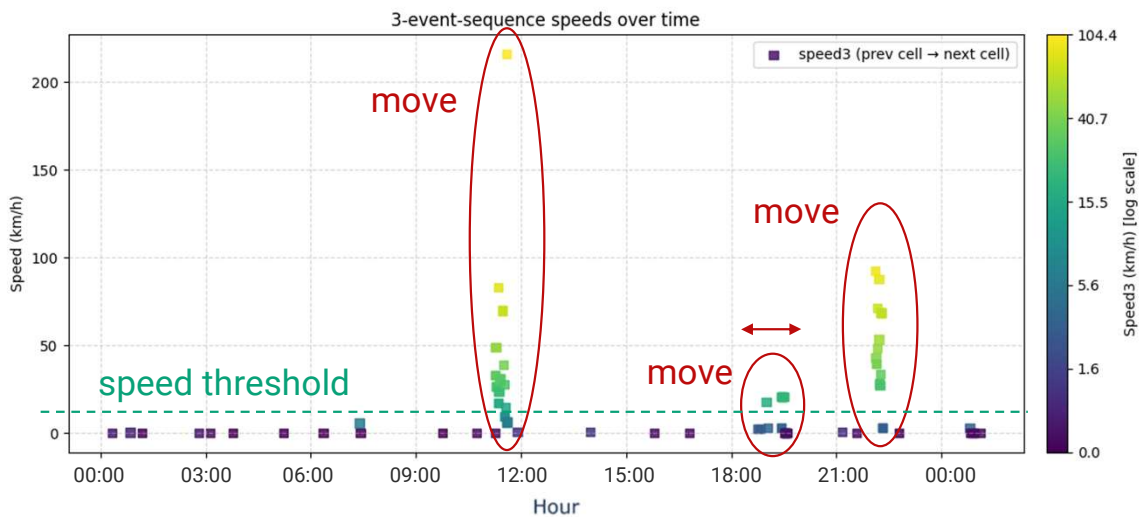
Speed-Based

- We repeat this calculation for **all event changes**.



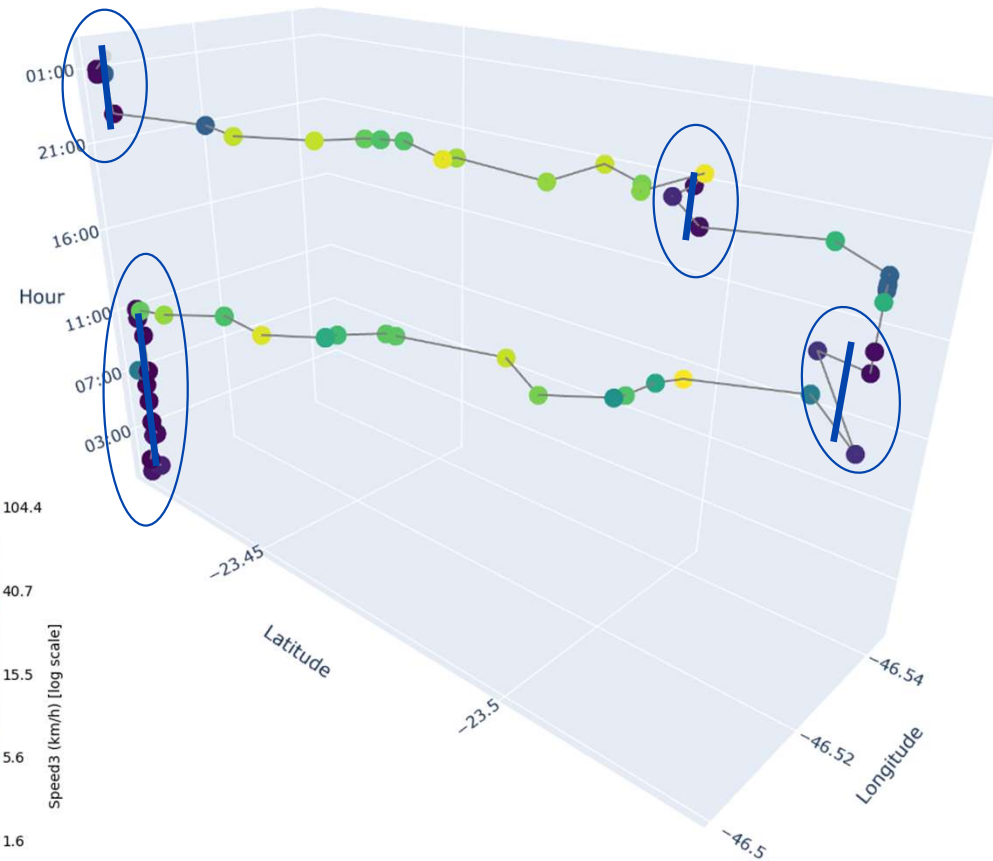
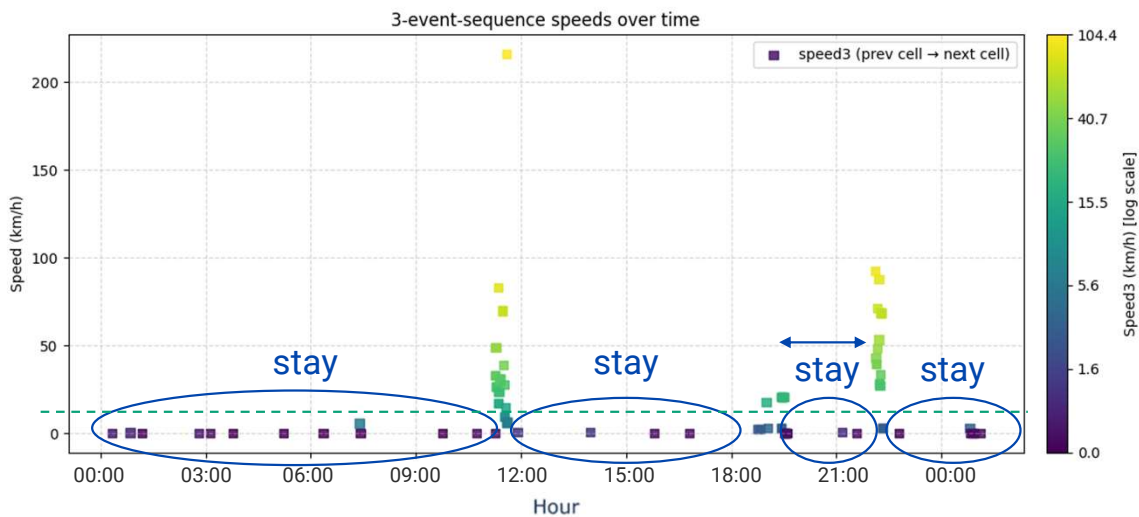
Speed-Based

- Sequences with speeds above the **threshold** (e.g. 5-10 km/h) are tagged as **moves**.
- Stays and/or moves shorter than a **minimum time** (e.g. 5-10 min) are often absorbed.



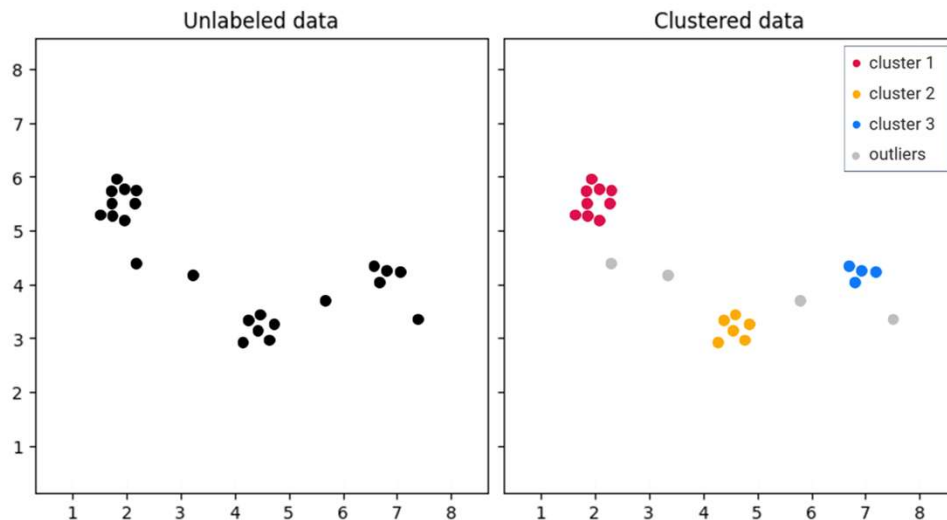
Speed-Based

- The **location** of each **stay** can be approximated as the (weighted) **centroid** of the “still” cells.



Clustering-Based

- What is **clustering**?
 - ML technique that **groups data points into natural splits or “clusters”** based on their similarity.
 - Elements in the same cluster are more similar to each other than to those in other clusters.



- **Key idea:** clustering events according to their location and time. Detected clusters correspond to stays, whereas outliers are tagged as moves.

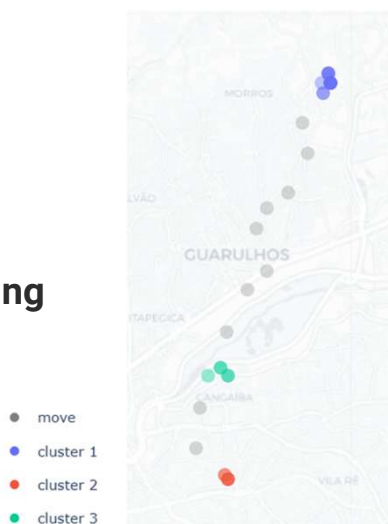


Clustering-Based: Implementation examples

DBSCAN clustering:

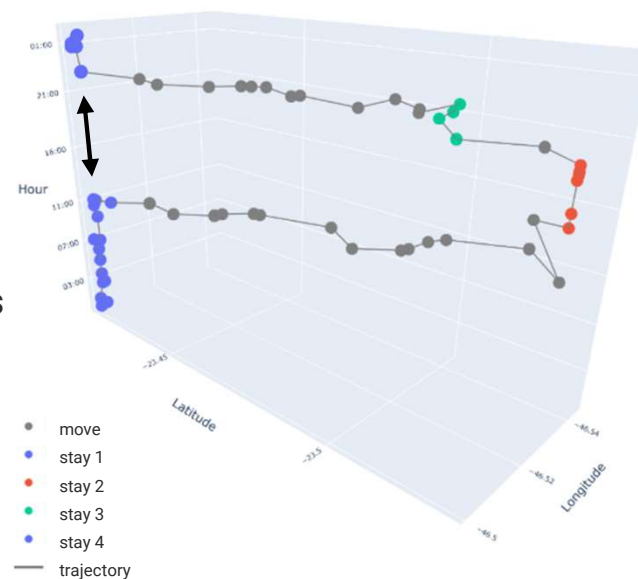
- Density-Based Spatial Clustering algorithms are able to group dense event clouds into location clusters (2D), which then can be post-processed into stays adding the time variable to the equation (3D).
- e.g.: [Evaluating the impact of cellular tower density on intercity travel OD identification \(Wang, L. et al, 2025\)](#)

2D clustering



+

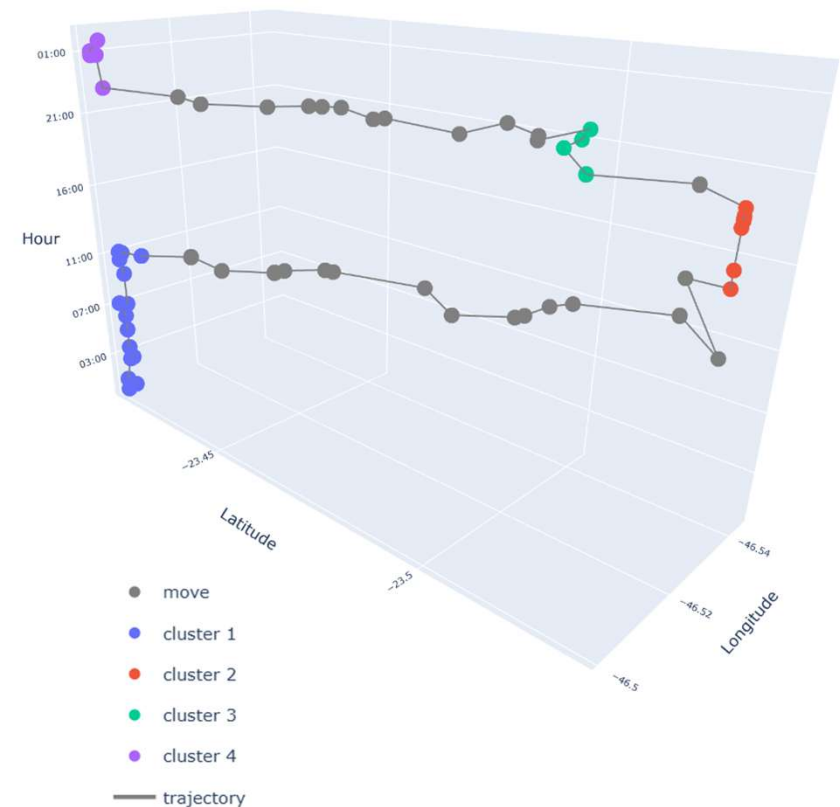
3D postprocess



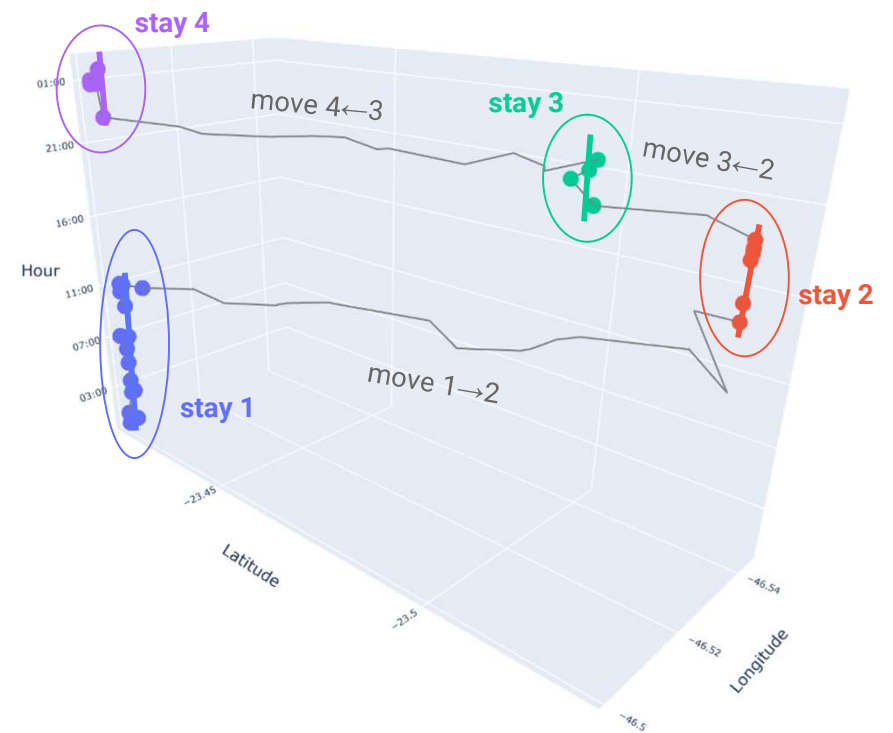
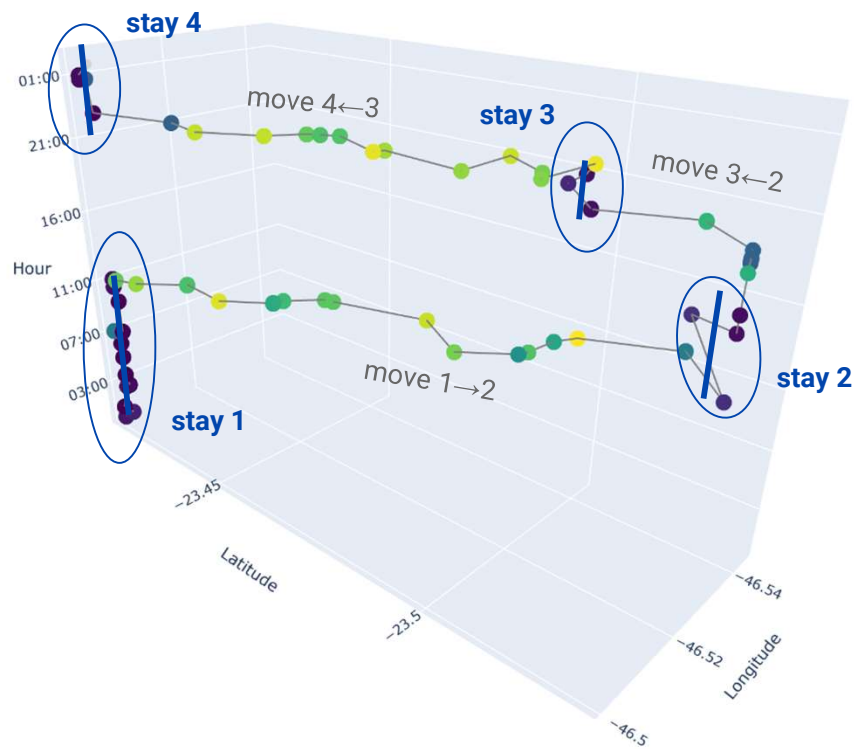
Clustering-Based: Implementation examples

Sliding window clustering:

- Scan through ordered cell points using a moving time–space window, grouping consecutive points that stay within a spatial radius for at least a minimum duration.
- e.g.:
 - (i) [scikit-mobility \(Python mobility analysis library\) “stay_locations” function](#),
 - (ii) [In Trajectory Data Mining: An Overview \(Yu Zheng, 2016\)](#)



Speed-Based vs Clustering-Based Results



MPD Continuity Models: Advanced - Reading Material

MPD Continuity Models: Advanced

- Determining the exact start/end time of each move/stay
 - Wang, F., & Chen, C. (2018). On data processing required to derive mobility patterns from passively-generated mobile phone data. Transportation Research Part C: Emerging Technologies, 87, 58–74. <https://doi.org/10.1016/j.trc.2017.12.003>
- Disaggregation of stays into stops and activities for long-distance trip inference
 - García-Albertos, P., Cantú Ros, O.G., Ciruelos C., Herranz, R.: Understanding door-to-door travel times from opportunistically collected mobile phone records, SESAR innovation days 2017. https://www.academia.edu/70130499/Understanding_Door_to_Door_Travel_Times_from_Opportunistically_Collected_Mobile_Phone_Records_A_Case_Study_of_Spanish_Airports
- Tour sequence inference (e.g. for tourism)
 - Chu, C.-P., & Chou, Y.-H. (2021). Using cellular data to analyze the tourists' trajectories for tourism destination attributes: A case study in Hualien, Taiwan. Journal of Transport Geography, 96, 103178. <https://doi.org/10.1016/j.jtrangeo.2021.103178>
- “Special” cell awareness: some cells (e.g. roads or metro cells) are prone to generating “move” events
 - Leontiadis, I., Lima, A., Kwak, H., Stanojevic, R., Wetherall, D., & Papagiannaki, K. (2014). From cells to streets: Estimating mobile paths with cellular-side data. Proceedings of the 10th ACM International Conference on Emerging Networking Experiments and Technologies (CoNEXT '14), 121–132. Association for Computing Machinery. <https://doi.org/10.1145/2674005.2674982>
- Abroad trips (with vs without Roaming Out data)
 - Burrieza-Galán, J., Jordá, R., Gregg, A., Ruiz, P., Rodríguez, R., Sala, M. J., Torres, J., García-Albertos, P., Cantú Ros, O. G., & Herranz, R. (2023). A methodology for understanding passenger flows combining mobile phone records and airport surveys: Application to Madrid-Barajas Airport after the COVID-19 outbreak. Journal of Air Transport Management, 110, 102370. <https://www.sciencedirect.com/science/article/abs/pii/S0969699721001447>

MPD Continuity Models: Practical Exercise

MPD Continuity Models: Intro to Practical Exercise

- **Task:** Detecting meaningful locations from a dataset.
- **Exercise Link:**
<https://github.com/Flowminder/WB-GDF-Modern-Data-Workflows-Code-Tasks>

The screenshot shows the GitHub repository page for 'WB-GDF-Modern-Data-Workflows-Code-Tasks'. The README file is open, displaying the repository's purpose and a table of course sessions. A red arrow points from the 'Exercise Link' in the list to the 'Session 2.4.1 Continuity models' row in the 'Course Sessions' table.

WB-GDF: Modern Data Workflows Code Tasks

This is a repository containing the coding exercises for the WB-GDF Course 2: Foundations in Mobile Phone Data (MPD) for Policy: Modern Data Workflows

Course Sessions

Session	Session Title	Link to Colab
Session 2.3.4	Input data quality assurance and cleaning	Open in Colab
Session 2.4.1	Continuity models	Open in Colab
Session 2.4.2	Meaningful Locations and Usual Environments	(to be added)
Session 2.4.3	Calculating aggregates	(to be added)
Session 2.5.1	Quality Assessment of Estimates for Decision-Making	(to be added)



The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).





Mobile Phone Data for Policy Programme (WB-GDF)
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